

SAURABH KASHID

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Work Experience

Joulea

Atlanta, GA

Founding Autonomy Engineer

Feb 2023 – Present

- Led cross functional teams, including contractors on Android app development (integrating DJI OSDK with MSDK for data transfer) and training drone pilots during field operations, flight tests, and system deployment.
- Developing LiDAR-camera extrinsic calibration pipeline using FAST-Calib for accurate cross sensor alignment.
- Calibrated camera intrinsics and lens distortion using OpenCV, achieving sub-pixel (0.7 px) reprojection error.
- Augmented the LiDAR-Inertial EKF with Visual Odometry measurements for robust, degeneracy-aware state estimation, improving pose accuracy in low-observability and GPS-denied environments.
- Developed GICP-based LiDAR-to-ENU alignment, enabling dynamic switching between local SLAM and GPS frames and improving global pose consistency across missions.
- Integrated Neo-D9S RTK correction with Ublox-F9R GNSS, enhancing localization accuracy by up to 2.5× and mitigating EMI induced positioning drift through optimized shielding.
- Implement real-time LiDAR obstacle detection on NVIDIA Orin using voxel filtering for safer navigation.
- Unified multi-session SLAM maps using TEASER++ dual-loss robust registration. voxelized maps for efficient storage and global relocalization, ensuring consistent large-scale mapping over 7M+ sq ft.
- Investigate STD loop closure with GTSAM pose graph optimization techniques to enhance lidar SLAM.
- Maintain CI pipeline with pre-commit and GoogleTest, ensuring faster and more reliable deployments.
- Executed over 300+ hours of autonomous drone inspection flights, covering more than 10 million sq. ft. of building surface area for high-resolution mapping, structural analysis, and data acquisition.
- Generated building models from point cloud data with Patchwork++ ground segmentation, optimizing code for a 30% reduction in processing time and validating improvements through Software-in-the-Loop (SIL) simulation and Hardware-in-Loop (HIL) to enable faster and more efficient coverage path planning.

EDUCATION

Worcester Polytechnic Institute

Jan 2022 – Dec 2023

Masters of Science, Robotics Engineering - [GPA - 3.77]

Worcester, MA

Savitribai Phule Pune University

Aug 2017 – May 2020

Bachelors of Engineering, Mechanical Engineering - [CGPA - 8.5]

Pune, Maharashtra, India

TECHNICAL SKILLS

Skills: Python, C++, Open3d, Blender, NetworkX, , Linux, Docker, Pytorch, OpenCV, Weights & Biases, MatLab, PCL, ROS, ROS2, Rviz, Shapely, Gazebo, GitHub, plotjuggler(ROS), Google OR-Tools

PROJECTS

3D Reconstruction using Structure from Motion (SfM)

Mar 2024 – Mar 2024

- Developed a complete Structure from Motion (SfM) pipeline from scratch, implementing core computer vision algorithms for robust 3D scene reconstruction and camera pose estimation.
- Utilized incremental SfM with non-linear optimization (Bundle Adjustment) to accurately triangulate 3D points and refine camera parameters.

Panoramic Image Stitching Pipeline

Feb 2024 – Feb 2024

- Integrated Panorama stitching with feature-matching + RANSAC in an OpenCV pipeline, using Poisson blending for seamless, artifact-free panoramas.
- Trained a HomographyNet from scratch on MS-COCO for direct homography regression.

6D Pose Detection using Dense Fusion on Linemode Dataset

Mar 2023 – Apr 2023

- Developed a 6D pose detection system using Dense Fusion on the modified Linemode Dataset.
- Applied state-of-the-art techniques on 6 classes out of 15 of linemode in deep learning and computer vision to improve the performance of the system.

Active Vision

Sep 2022 – Nov 2022

- Segmented point cloud objects from the major plane using the RANSAC algorithm in ROS2 Gazebo
- Combined point clouds using ICP and estimated grasp points using grasp approximation methods in C++.

CERTIFICATIONS

• FAA Part 107 Remote Pilot License

31 Mar 2023